

Raj Kumar Mahto

+91-9599201959 — rajmahto9876@gmail.com

www.linkedin.com/in/raj-kumar-mahto-b78069127 — https://github.com/rajmahto9876

https://rajmahto9876.github.io/Blog/resume/

Professional Summary

Embedded Software Engineer with 5+ years of experience in Embedded C/C++, firmware development, SoC bring-up, and embedded Linux systems for automotive and IoT platforms. Skilled in ARM Cortex-based systems, peripheral driver development, RTOS, and low-level debugging using GDB, JTAG, and Lauterbach Trace32. Experienced in SPI, I2C, UART, DMA, Linux kernel modules, hardware interfacing, and embedded system validation.

Education

National Institute of Technology, Warangal

PG Diploma (M.Tech) in Control and Automation

CGPA: 8.43

Maharaja Agrasen Institute of Technology, Delhi

B.Tech in Electrical and Electronics Engineering

CGPA: 7.99

Technical Skills

Programming Languages: Embedded C, Embedded C++, Python

MCU / SoCs: ARM Cortex-M, ARM Cortex-A, NXP S32K5xx, ESP32, STM32, Basic Risc-V.

Protocols & Networking: UART, SPI, I2C, DMA, MQTT, UDP, HTTPS, Wi-Fi (ESP32)

Operating Systems: FreeRTOS, Linux

Embedded Linux: Linux Kernel Modules, Character Device Drivers, Memory-Mapped I/O (MMIO), Cross Compilation, Kernel Debugging, User-Kernel Communication

Build & Toolchains: CMake, LLVM, Makefiles, QEMU, Keil IDE, MPLABX

Debug & Validation Tools: Lauterbach Trace32, Veloce Emulation, GDB, Valgrind

Version Control: Git, Bitbucket, code reviews, collaborative development workflows

Hardware tools: Oscilloscope, Multimeter, Logic Analyzer, ST-Link, ULINK

Code Quality & Compliance: MISRA-C coding guidelines, Static code analysis.

Professional Experience

VVDN Technologies (Onsite: NXP Semiconductors)

May 2024 – July 2025

Senior Software Engineer

- Designed, developed, tested, and debugged embedded firmware for ARM-based automotive SoCs including peripheral validation, interrupt verification, timer subsystem testing, and low-level register debugging. Worked on platform initialization, startup code, linker configuration, and subsystem bring-up for automotive SoC platforms.
- Performed pre-silicon validation on the Veloce emulation platform, validated 8+ IP blocks some of them are interrupt controllers, timers, clock modules, and reset logic. Utilized Lauterbach Trace32 (T32) for register-level debugging, instruction tracing.
- Automated testing reduced regression time by 20%. Performed root-cause analysis and low-level debugging of complex hardware-firmware interaction issues using Trace32, GDB, and JTAG.

IoTfy Solutions Pvt. Ltd., Delhi

May 2021 – May 2024

Tech Lead – Embedded Systems

- Developed Linux kernel modules and user-space applications for hardware interaction using memory-mapped I/O (MMIO) and device interfaces.

- Mentored and guided a team of embedded engineers, enforcing coding standards, design reviews, and best practices.
- Designed **FreeRTOS-based task** architecture using queues, semaphores, mutexes, and software timers.

Copper Connection Pvt. Ltd., Delhi
Embedded Engineer

April 2020 – May 2021

- Worked on embedded firmware development and feature implementation for PBX systems.
- Supported hardware prototyping, debugging, and system validation activities.

Key Projects

Linux Device Driver Project

June 2025 – April 2026

- Developed a **Linux kernel driver** for the TM1637 7-segment display on BeagleBone Black using GPIO bit-banging and kernel-space APIs.
- Developed **Linux platform drivers** with Device Tree integration, implemented probe/remove callbacks, and utilized **devm-managed resources** for efficient driver lifecycle management.
- Performed **kernel debugging and root-cause analysis** using KGDB, printk, stack trace inspection, and memory fault analysis on ARM-based Linux systems.
- Developed and debugged applications interfacing with **GPIO, UART, SPI, and I2C peripherals**, including interrupt handling and hardware-software integration.

Hardware Security Engine (HSE) – Pre-Silicon Project

May 2024 – June 2025

- Developed and maintained **CMake build scripts** for compiling bare-metal firmware in a multi-module environment. Worked on low-level firmware development and debugging for RISC-V based embedded platforms.
- Worked extensively on **linker scripts and startup files** for system initialization and memory mapping. Enabled and validated critical **bare-metal security and system IPs** including Interrupt, Timer, Clock Subsystems, MC_RGM, and MC_ME.
- Implemented and executed **unit and system-level test cases** to verify firmware correctness and IP behavior. Performed **system testing in a pre-silicon environment** using the Veloce emulation platform.

IoT Switches, Plugs, Wi-Fi Mesh Network Implementation (ESP32 & Chimney Systems)

May 2023 – May 2024

- Designed and implemented a Wi-Fi Mesh network supporting 40+ devices under a single IP.
- Designed and developed an **IoT-enabled smart devices** with mobile app, HTTP, web server, UDP, MQTT, and cloud integration.
- Strong experience in bare-metal firmware, RTOS-based embedded systems, and low-level hardware interfacing.
- Enabled real-time monitoring and remote control through cloud-connected embedded firmware.

BLDC Fan Development

May 2021 – May 2024

- Optimized BLDC fan tuning and wattage control algorithms to achieve a **5-star efficiency rating**.
- Assisted in hardware design activities including **PCB layout** and **SMPS integration**. Implemented **low-voltage and high-voltage cut-off protection circuits** to enhance system safety and reliability.
- Developed **multi-mode control functionality**, supporting IR remote, RF remote, and cloud-based operation.